Oxford Nanopore Technologies

Rapid sequencing DNA V14 - barcoding (SQK-RBK114.24 or SQK-RBK114.96)

Overview

Before beginning, we strongly recommend that first-time users read and review the complete online protocol on the Nanopore Community. The online protocol provides the full instructions, additional commentary, and considerations for various steps, as well as guidance on input amounts, DNA handling, and sequencing. **This reference card is designed for experienced users.**

1. Library preparation (~45 minutes)

1 Program the thermal cycler: 30°C for 2 minutes, then 80°C for 2 minutes.

2 Thaw kit components at room temperature, spin down briefly using a microfuge and mix by pipetting as indicated by the Table below:

Reagent	1. Thaw at room temperature	2. Briefly spin down	3. Mix well by pipetting
Rapid Barcodes (RB01-24 or RB01-96)	Not frozen	✓	✓
Rapid Adapter (RA)	Not frozen	✓	✓
AMPure XP Beads (AXP)	✓	✓	Mix by pipetting or vortexing immediately before use
Elution Buffer (EB)	✓	✓	✓
Adapter Buffer (ADB)	✓	✓	Mix by vortexing

3 Prepare the DNA in nuclease-free water.

1. Transfer 200 ng of genomic DNA per sample into 0.2 ml thin-walled PCR tubes or an Eppendorf twin.tec® PCR plate 96 LoBind.
2. Adjust the volume of each sample to 10 µl with nuclease-free water.
3. Pipette mix the content of the tubes 10-15 times to avoid unwanted shearing.
4. Spin down briefly in a microfuge.

4 In the 0.2 ml thin-walled PCR tubes or an Eppendorf twin.tec® PCR plate 96 LoBind, mix the following:

Reagent	Volume per sample
Template DNA (200 ng from previous step)	10 µl
Rapid Barcodes (RB01-24 or RB01-96, one for each sample)	1.5 µl
Total	**11.5 µl**

5. Ensure the components are thoroughly mixed by pipetting and spin down briefly.

6. Incubate the tubes or plate at 30°C for 2 minutes and then at 80°C for 2 minutes. Briefly put the tubes or plate on ice to cool.

7. Spin down the tubes or plate to collect the liquid at the bottom.

8. Pool all barcoded samples in a clean 2 ml Eppendorf DNA LoBind tube, noting the total volume.

	Volume per sample	For 4 samples	For 12 samples	For 24 samples	For 48 samples	For 96 samples
Total volume	11.5 µl	46 µl	138 µl	276 µl	552 µl	1,104 µl

9. Resuspend the AMPure XP Beads (AXP) by vortexing.

10. Add an equal volume of resuspended AMPure XP Beads (AXP) to the entire pooled barcoded sample, and mix by flicking the tube.

.	Volume per sample	For 4 samples	For 12 samples	For 24 samples	For 48 samples	For 96 samples
Volume of AMPure XP Beads (AXP) added	11.5 µl	46 µl	138 µl	276 µl	552 µl	1,000 µl

11. Incubate on a Hula mixer (rotator mixer) for 10 minutes at room temperature.

12. Prepare at least 2 ml of fresh 80% ethanol in nuclease-free water.

13. Spin down the sample and pellet on a magnet. Keep the tube on the magnet, and pipette off the supernatant.

14. Keep the tube on the magnet and wash the beads with 1 ml of freshly prepared 80% ethanol without disturbing the pellet. Remove the ethanol using a pipette and discard.

15. Repeat the previous step.

16. Briefly spin down and place the tube back on the magnet. Pipette off any residual ethanol. Allow to dry for 30 seconds, but do not dry the pellet to the point of cracking.

17. Remove the tube from the magnetic rack and resuspend the pellet in 15 µl Elution Buffer (EB) per 24 barcodes used.

.	For 01-24 barcodes	For 48 barcodes	For 72 barcodes	For 96 barcodes
Volume of Elution Buffer (EB)	15 µl	30 µl	45 µl	60 µl

Note: Always elute your sample in at least 15 µl of Elution Buffer (EB). The elution volume can be adjusted for over 24 barcodes by following the Table.

18. Incubate for 10 minutes at room temperature.

19. Pellet the beads on a magnet for at least 1 minute, until the eluate is clear and colourless.

20. Remove and retain the full volume of eluate into a clean 1.5 ml Eppendorf DNA LoBind tube.

- Remove and retain the eluate which contains the DNA library in a clean 1.5 ml Eppendorf DNA LoBind tube.

- Dispose of the pelleted beads.

21. Transfer 11 µl of the sample into a clean 1.5 ml Eppendorf DNA LoBind tube.

22. In a fresh 1.5 ml Eppendorf DNA LoBind tube, dilute the Rapid Adapter (RA) as follows and pipette mix:

Reagent	Volume
Rapid Adapter (RA)	1.5 µl
Adapter Buffer (ADB)	3.5 µl
Total	**5 µl**

23. Add 1 µl of the diluted Rapid Adapter (RA) to the barcoded DNA.

24. Mix gently by flicking the tube, and spin down.

25. Incubate the reaction for 5 minutes at room temperature.

Tip: While this incubation step is taking place you can proceed to the Flow Cell priming and loading section of the protocol.

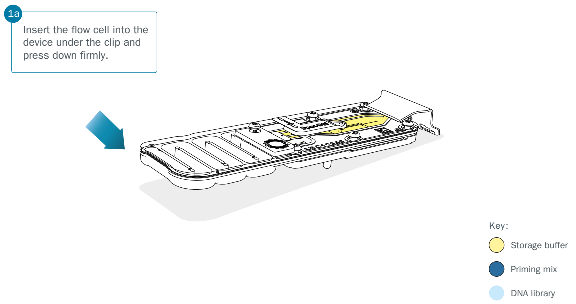
2. Priming and loading the MinION and GridION Flow Cell (~10 minutes)

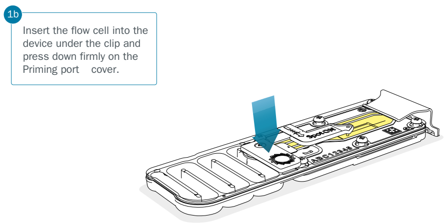
1. Thaw the Sequencing Buffer (SB), Library Beads (LIB) or Library Solution (LIS, if using), Flow Cell Tether (FCT) and Flow Cell Flush (FCF) at room temperature before mixing by vortexing. Then spin down and store on ice.

2. To prepare the flow cell priming mix with BSA, combine the following reagents in a fresh 1.5 ml Eppendorf DNA LoBind tube. Mix by inverting the tube and pipette mix at room temperature:

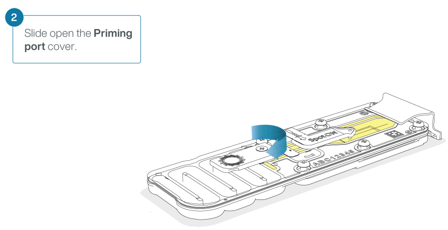
Reagents	Volume per flow cell
Flow Cell Flush (FCF)	1,170 µl
Bovine Serum Albumin (BSA) at 50 mg/ml	5 µl
Flow Cell Tether (FCT)	30 µl
Final total volume in tube	**1,205 µl**

3. Open the MinION or GridION device lid and slide the flow cell under the clip. Press down firmly on the priming port cover to ensure correct thermal and electrical contact.





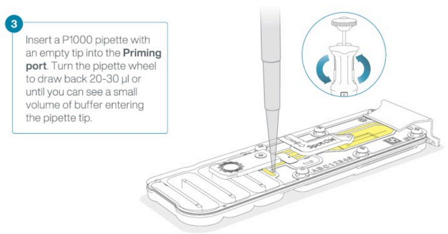
4. Slide the flow cell priming port cover clockwise to open the priming port.



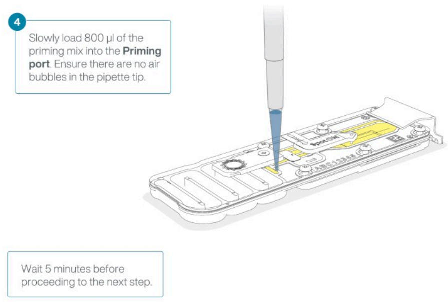
5 After opening the priming port, check for a small air bubble under the cover. Draw back a small volume to remove any bubbles:

1. Set a P1000 pipette to 200 µl
2. Insert the tip into the priming port
3. Turn the wheel until the dial shows 220-230 µl, to draw back 20-30 µl, or until you can see a small volume of buffer entering the pipette tip

Note: Visually check that there is continuous buffer from the priming port across the sensor array.



6 Load 800 µl of the priming mix into the flow cell via the priming port, avoiding the introduction of air bubbles. Wait for five minutes. During this time, prepare the library for loading by following the steps below.



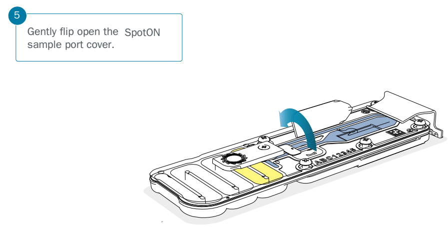
7 Thoroughly mix the contents of the Library Beads (LIB) by pipetting.

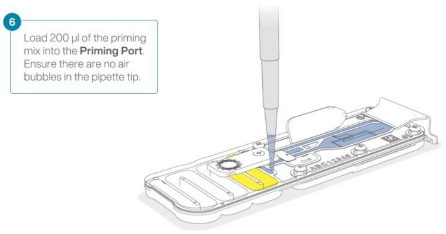
8 In a new 1.5 ml Eppendorf DNA LoBind tube, prepare the library for loading as follows:

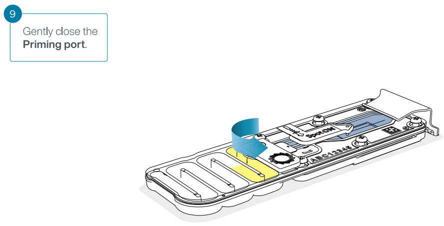
Reagent	**Volume per flow cell**
Sequencing Buffer (SB)	37.5 µl
Library Beads (LIB) mixed immediately before use, or Library Solution (LIS), if using	25.5 µl
DNA library	12 µl
Total	**75 µl**

9 Complete the flow cell priming:

1. Gently lift the SpotON sample port cover to make the SpotON sample port accessible.
2. Load **200 µl** of the priming mix into the flow cell priming port (**not** the SpotON sample port), avoiding the introduction of air bubbles.

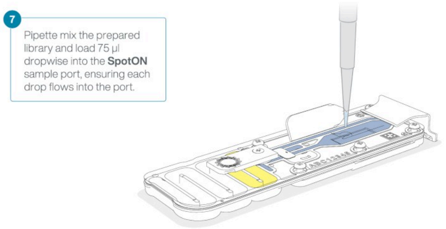




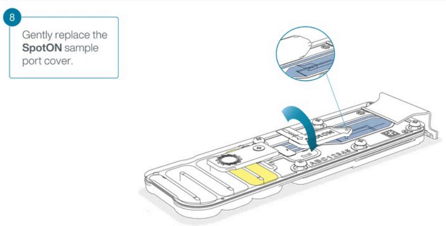


10. Mix the prepared library gently by pipetting up and down just prior to loading.

11. Add 75 µl of the prepared library to the flow cell via the SpotON sample port in a dropwise fashion. Ensure each drop flows into the port before adding the next.



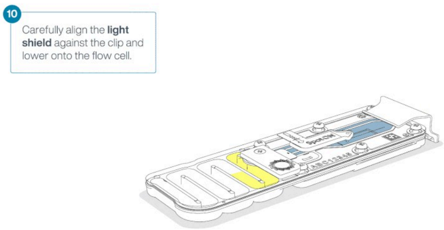
12. Gently replace the SpotON sample port cover, making sure the bung enters the SpotON port and close the priming port.



13. Place the light shield onto the flow cell, as follows:

1. Carefully place the leading edge of the light shield against the clip.
   Note: Do not force the light shield underneath the clip.

2. Gently lower the light shield onto the flow cell. The light shield should sit around the SpotON cover, covering the entire top section of the flow cell.



3. Flow cell reuse and returns (~0 minutes)

1. After your sequencing experiment is complete, if you would like to reuse the flow cell, please follow the Flow Cell Wash Kit protocol and store the washed flow cell at +2°C to +8°C.

The Flow Cell Wash Kit protocol is available on the Nanopore Community.

2. Alternatively, follow the returns procedure to send the flow cell back to Oxford Nanopore.

Instructions for returning flow cells can be found here.